

Distributional Reinforcement Learning

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1 Necessitation

Among all machine learning paradigms, **Reinforcement Learning** is perhaps the closest to natural learning mechanisms seen in animals, including humans. It involves the interaction of an agent with an ambient space, the **environment**, through which it learns how to behave optimally in that setting.

Classical reinforcement learning is primarily concerned with learning *strategies* through the estimation of expectations of cumulative future reward. More precisely, the central object is typically a value function that assigns to each state, or state-action pair, the expected long-term payoff under a given policy or under an optimal decision rule. This paradigm has proved remarkably successful across a wide range of sequential decision-making tasks, especially in settings where the environment is only partially known or must be learned through interaction.

Despite its success, the classical framework compresses all randomness in future outcomes into a single scalar quantity - the expectation. As a consequence, potentially important information about the variability, asymmetry, tail behavior, or multimodality of returns is discarded. Two distinct return mechanisms may therefore appear identical from the viewpoint of expected value, even though they induce very different levels of risk or uncertainty. Moreover, mean and variance even when accounted for jointly, only enables the estimation of bounds, e.g through the Markov/Chebyshev inequalities for the right tail, Cantelli's inequality for the left tail, but does not allow the explicit enumeration of the tail events.

Several applications require the need to consider higher moments of a random variable. Investment decisions require weighing expected profit against volatility, Lottery tickets offer negative expected returns yet attract buyers, and during a snowstorm, relying on the average bus arrival frequency leads to disappointment

2 Introduction

Just as a color photograph conveys more information than a black-and-white one, the return distribution contains more information than the expected return alone. The expected return is a scalar, whereas the return distribution is an infinite-dimensional object capturing the full range of possible outcomes. Moving from scalars to distributions fundamentally changes the analytical setting of reinforcement learning.

2.1 The distributional shift

Shift in Planning - The standard Bellman Equations are usually denoted using value functions as

$$V_t^\pi(s) = \mathbb{E} [R + \gamma V_{t+1}^\pi(S_{t+1}) \mid S_t = s] \quad (1)$$

While the recursive structure of dynamic programming is preserved, the Bellman equation is lifted to a *distributional Bellman equation*, and Bellman operators become operators on probability measures

$$G_t^\pi(s) \stackrel{\mathcal{D}}{=} R + \gamma G_{t+1}^\pi(S_{t+1}), \quad S_t = s.$$

where $G_t^\pi(s), G_{t+1}^\pi(S_{t+1}), R, S_{t+1}$ are all random variables. This transition replaces scalar-valued functions with distribution-valued mappings and necessitates the use of probability metrics to study convergence. Crucially, the distributional Bellman operator is no longer a contraction under

standard norms, and its behavior depends fundamentally on the choice of metric (e.g., Wasserstein distances).

Learning - On the tabular side, Monte Carlo methods naturally extend to learning return distributions. Temporal-difference learning, however, requires a careful reformulation the recursive decomposition now acts on distributions, and practical algorithms must incorporate projection operators to maintain tractable representations. In function approximation settings, additional challenges arise. Methods such as *categorical distributional reinforcement learning* [1] approximate return distributions using discrete supports, leading to algorithms such as categorical temporal-difference learning. Alternatively, quantile-based methods represent distributions through their inverse cumulative functions. A key conceptual shift is that errors between estimates must now be measured using probability metrics rather than point-wise differences. This introduces a geometric structure on the space of return distributions, where operators involve compositions of translation, scaling, and expectation, reflecting the underlying stochastic dynamics.

Risk-sensitive behavior - Knowledge of the return distribution allows agents to pursue *risk-sensitive* reinforcement learning — for instance, avoiding states with a high probability of failure or penalizing high-variance decisions. This can make behavior more robust and better suited to real-world applications.

Representation - Distributional reinforcement learning must approximate a full return distribution using only a finite number of parameters, which creates practical computational challenges. Two major approximation schemes exist - *categorical representations*, which fix a uniform grid over possible returns and learn the probability assigned to each bin, and *quantile representations*, which instead learn the positions of a fixed number of equally weighted particles. Because these representations capture different structural properties of the return distribution, they differ in both approximation quality and computational convenience, making the choice of representation an important design decision in practice.

2.2 Results

The main ideas presented are -

1. distributional decomposition and relation to classical Bellman equations and the derivation of the distributional Bellman operator from an analytic perspective
2. effects of variance and the higher moments
3. learning from distributions, the proposal of categorical temporal difference learning, and how to handle the size of the distributions before enabling algorithms to solve the problem at hand and projected dynamic programming
4. distributional reinforcement learning from a function approximation (deep learning) perspective

3 Mathematical Modelling

3.1 Distributions

There are events when the return/reward distributions are discrete, continuous with densities, or fall into neither category. This is because a random variable, if continuous, may still not have a density, for example, *singular distributions* which are concentrated on a set of Lebesgue measure 0. As such, the obvious choice is to work with *cumulative distributions* (CDFs).

3.1.1 Cumulative Distribution Functions

Let Z be a real-valued random variable with distribution $\mu \in \Delta(\mathbb{R})$. Its cumulative distribution function (CDF) is defined by

$$F_Z(x) := \mathbb{P}(Z \leq x), \quad x \in \mathbb{R}.$$

The CDF F_Z is non-decreasing, right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F_Z(x) = 0, \quad \lim_{x \rightarrow \infty} F_Z(x) = 1,$$

thus providing a complete characterization of the law of Z .

It is convenient to regard CDFs as elements of an ambient topological vector space of real-valued functions on $\mathbb{R}$, for instance $L^\infty(\mathbb{R})$, while noting that the collection of valid CDFs forms a nonlinear subset of this space. In this setting, translations act through bijective shift operators $(T_a F)(x) := F(x - a)$, $a \in \mathbb{R}$. Each T_a is invertible with inverse T_{-a} , and hence is an automorphism of the ambient space. When restricted to the class of CDFs, it acts as a homeomorphism onto its image and preserves the probabilistic structure defined by the CDF. [8] Equivalently, if one defines the dilation operator

$$(D_b F)(x) := F\left(\frac{x}{b}\right), \quad b > 0,$$

More generally for any affine transformation $Z' = a + bZ$ with $b > 0$, the induced action on CDFs is

$$F_{Z'}(x) = F_Z\left(\frac{x - a}{b}\right).$$

then $F_{Z'} = T_a D_b F_Z$. Thus, affine transformations of random variables correspond to compositions of translation and dilation operators on the ambient function space. ¹

3.2 Markov Decision Processes

An MDP is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, \mathcal{R}, \gamma)$, where $\mathcal{S}$ and $\mathcal{A}$ denote the state and action spaces, $P(\cdot | s, a)$ is the transition kernel, $\mathcal{R}$ is the reward space, and $\gamma \in [0, 1)$ is the discount factor. A stochastic process $(S_t)_{t=1}^T$ is said to be *Markovian* if the conditional distribution of the next state depends only on the current state and action, and not on the past history. That is,

$$\mathbb{P}(S_{t+1} | S_1, \dots, S_t, A_1, \dots, A_t) = \mathbb{P}(S_{t+1} | S_t, A_t). \quad (2)$$

3.3 Classical Bellman Equations and Dynamic Programming

Bellman Decomposition - The classical formulation operates on the *expectation* of the return. Define the value function

$$V_t^\pi(s) := \mathbb{E}[G_t^\pi | S_t = s]. \quad (3)$$

Using the one-step decomposition

$$G_t^\pi \stackrel{d}{=} R(S_t, A_t) + \gamma G_{t+1}^\pi, \quad (4)$$

¹This perspective provides the additional benefit of identifying a structural backbone which is useful in distributional dynamic programming, where Bellman recursions act on distributions through precisely such shift, scaling, and averaging operations.

and taking expectations on both sides, we obtain

$$V_t^\pi(s) = \mathbb{E}^\pi [R(S_t, A_t) + \gamma G_{t+1}^\pi \mid S_t = s] = \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} P(s'|s, a) [r(s, a) + \gamma V_{t+1}^\pi(s')] \quad (5)$$

$$V_t^\pi = r^\pi + \gamma P^\pi V_{t+1}^\pi, \quad (6)$$

where the preceding inequality is the vector form with

$$r^\pi(s) := \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) r(s, a), \quad (P^\pi V)(s) := \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) V(s'). \quad (7)$$

This is the classical dynamic program under the classical operator. Depending on utility, one may present the effect of the policy-Bellman Operator or the optimal Bellman operator.

Policy Evaluation - For a fixed policy π , the value function satisfies

$$V_t^\pi(s) = \mathbb{E}^\pi [R + \gamma V_{t+1}^\pi(S_{t+1}) \mid S_t = s], \quad V^\pi = \mathcal{B}^\pi V^\pi \quad (8)$$

Bellman Optimality Equation - The optimal value function satisfies

$$V_t^*(s) = \sup_{a \in \mathcal{A}} \mathbb{E} [R + \gamma V_{t+1}^*(S_{t+1}) \mid S_t = s, A_t = a], \quad V^* = \mathcal{B}^* V^* \quad (9)$$

Dynamic Programming - For the finite-horizon scenario, the standard approach is *backward recursion* as follows -

$$\text{Set } V_{T+1}(s) = 0 \quad (10)$$

$$\text{Then } \forall t \in T, \dots, 1 \quad (11)$$

$$Q_t^*(s, a) = \mathbb{E} [R + \gamma V_{t+1}^*(S_{t+1}) \mid S_t = s, A_t = a], \Rightarrow V_t^*(s) = \max_{a \in \mathcal{A}} Q_t^*(s, a) \quad (12)$$

$$\pi_t^*(s) \in \arg \max_{a \in \mathcal{A}} Q_t^*(s, a) \quad (13)$$

3.4 Distributional Dynamic Programming

Distributional Decomposition - Eliminating expectations actually now renders the distributional decomposition a simple one-step, discounted time-index separation.

Under Markovian states, the return may be expressed in a backward-recursive form with respect to the state variable using translational-invariance with respect to time

$$G_t^\pi(s) \stackrel{d}{=} R(s, a) + \gamma G_t^\pi(s'), \quad a \sim \pi(\cdot|s), \quad s' \sim P(\cdot|s, a). \quad (14)$$

3.5 Distributional Bellman Equation

In the distributional setting, one studies the law of the random return rather than only its expectation. For a fixed policy π , the random return satisfies the distributional Bellman equation

$$G_t^\pi \stackrel{d}{=} R_t + \gamma G_{t+1}^\pi, \quad (15)$$

where, conditional on (S_t, A_t) , the next state is distributed according to $P(\cdot \mid S_t, A_t)$ and the immediate reward is taken to be $r(S_t, A_t)$. Passing from random variables to cumulative distribution functions yields a dynamic programming recursion directly on CDFs.

Theorem 1. Consider $F_t^\pi(x \mid s) := \mathbb{P}(G_t^\pi \leq x \mid S_t = s)$, $x \in \mathbb{R}$. Then,

$$F_t^\pi(x \mid s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, a) F_{t+1}^\pi\left(\frac{x - r(s, a)}{\gamma} \mid s'\right). \quad (16)$$

The argument of the successor CDF is obtained by an affine transformation - a translation by the immediate reward and a scaling by the discount factor.

Proof. Fix t , $s \in \mathcal{S}$, and $x \in \mathbb{R}$. By definition,

$$F_t^\pi(x \mid s) = \mathbb{P}(G_t^\pi \leq x \mid S_t = s).$$

Using the one-step decomposition of the return, condition on the first action $A_t = a$ and next state $S_{t+1} = s'$ to obtain

$$F_t^\pi(x \mid s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \mathbb{P}(r(s, a) + \gamma G_{t+1}^\pi \leq x \mid S_{t+1} = s').$$

$$\text{Since } \gamma > 0, \quad r(s, a) + \gamma G_{t+1}^\pi \leq x \iff G_{t+1}^\pi \leq \frac{x - r(s, a)}{\gamma}.$$

$$\mathbb{P}(r(s, a) + \gamma G_{t+1}^\pi \leq x \mid S_{t+1} = s') = F_{t+1}^\pi\left(\frac{x - r(s, a)}{\gamma} \mid s'\right).$$

Substituting this into the previous expression yields (16). Now to express the decomposition and the recursion with respect to the distribution of returns, define a conditional *law* of the random returns under a policy π .² $\square$

Theorem 2. For any Borel subset $B \subseteq \mathbb{R}$, let $\eta_t^\pi(s)$ denote the law of G_t^π conditional on $S_t = s$, i.e.,

$$\eta_t^\pi(s)(B) := \mathbb{P}(G_t^\pi \in B \mid S_t = s), \quad B \subseteq \mathbb{R} \text{ Borel}. \quad (17)$$

Then,

$$\eta_t^\pi(s)(B) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \eta_{t+1}^\pi(s')\left(\frac{B - r(s, a)}{\gamma}\right), \text{ where } \frac{B - r(s, a)}{\gamma} := \{u \in \mathbb{R} : r(s, a) + \gamma u \in B\} \quad (18)$$

Proof. Fix $s \in \mathcal{S}$ and let $B \subseteq \mathbb{R}$ be Borel. By definition,

$$\eta_t^\pi(s)(B) = \mathbb{P}(G_t^\pi \in B \mid S_t = s). \quad (19)$$

Using the one-step decomposition $G_t^\pi \stackrel{d}{=} r(S_t, A_t) + \gamma G_{t+1}^\pi$, and conditioning on $A_t = a$ and $S_{t+1} = s'$, we obtain

$$\eta_t^\pi(s)(B) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \mathbb{P}(r(s, a) + \gamma G_{t+1}^\pi \in B \mid S_{t+1} = s') \quad (20)$$

²The *law* of a random variable is the probability measure induced by it. All random variables have an associated *law*. If the random variable has a distribution, then its law and distribution are synonymous

Now

$$r(s, a) + \gamma G_{t+1}^\pi \in B \iff G_{t+1}^\pi \in \frac{B - r(s, a)}{\gamma} \Rightarrow \mathbb{P}(r(s, a) + \gamma G_{t+1}^\pi \in B \mid S_{t+1} = s') = \eta_{t+1}^\pi(s') \left(\frac{B - r(s, a)}{\gamma} \right) \quad (21)$$

Substituting this into the previous expression yields (18). [1] uses a generalized notion of translation induced by the action of the pushforward $b_{R,\gamma}$ acting on the law as $b_{(R,\gamma)\#}\eta^\pi(\cdot)$. To establish notational equivalence, we use our earlier composite notation $T_R \circ D_\gamma$ or simply, $T_R D_\gamma$ $\square$

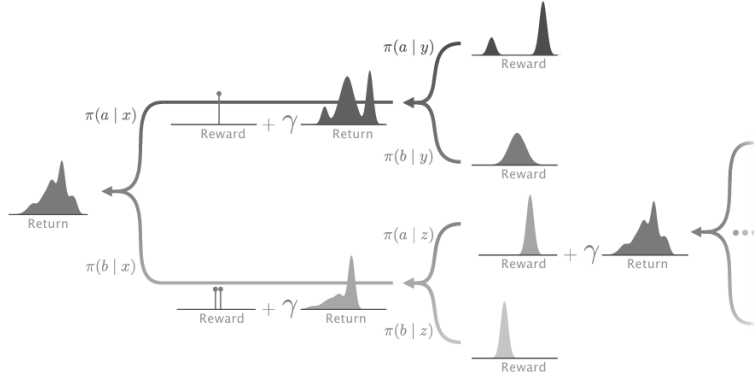


Figure 1: [1] - Sequential Translation Diagram

This, however, poses a key difficulty - the Bellman operator now acts on probability measures rather than scalar value functions, which introduces nonlinearity, loss of contraction under standard norms, and difficulty in propagating moments such as variance. In the next section, the propagation of higher moments has been illustrated through a formal procedure using the variance as in [10]. Later, we shall address the contraction properties and present the proposal in [1] of using the Wasserstein-2 distance as the contraction metric.

3.6 Distributional Bellman Operator

The *distributional Bellman operator* was implicitly constructed within the preceding derivation. The shift from value functions to distributions means that the distributional Bellman operator acts on the *law* of the distribution ³ Now we provide a formal definition Let $\mathcal{T}$ be the distributional Bellman operator. Then $\mathcal{T}$ satisfies

$$(\mathcal{T}^\pi G_t^\pi)(s) \stackrel{\mathcal{D}}{=} R + \gamma G_{t+1}^\pi(S_{t+1}) \quad (22)$$

The proof follows immediately from the classical analogue Thus $\mathcal{T}^\pi$ maps the random variable G evaluated at state $S_t = s$ to a new collection of random variables $R + \gamma G(S_{t+1})$. One drawback that becomes clearly evident from the proposition above is that $\mathcal{T}$ acts on a random variable, as such it maps onto the probability distribution of $(\mathcal{T}^\pi G_t^\pi)$. While it is possible to construct a space of random variables for the mapping to relate, that would requires further analytic considerations,

³The central object for any DP-style recursion is a **target**. The target is allowed to be any entity that sufficiently characterizes the objective of the problem - in the classical setting this was the value function, here it is the law, but it really can be any function

for simplicity, we consider the mapping from some sample space to the real line.

Proposition 1. *If $\eta_t : \mathcal{S} \rightarrow \mathcal{P}(\mathbb{R})$, assigns to each state s a probability law over returns at time t , then the policy-evaluation distributional Bellman operator is defined by*

$$(\mathcal{T}^\pi \eta_t)(s) = \mathbb{E}_\pi[(T_R D_\gamma)_\# \eta_{t+1}(S_{t+1}) \mid S_t = s] \iff \eta^\pi = \mathcal{T}^\pi \eta^\pi \quad (23)$$

Proof. The proof follows immediately from the definition of the distributional Bellman operator and (20) with the classical analogue

$$(\mathcal{B}^\pi V^\pi)(s) = \mathbb{E}_\pi[(T_R D_\gamma)_\# V_{t+1}^\pi(S_{t+1}) \mid S_t = s] \iff V^\pi = \mathcal{B}^\pi V^\pi \quad (24)$$

Uniqueness of fixed-points are guaranteed by the property of distances in metric spaces $d(x, y) = 0 \Rightarrow x = y$ formally through the contraction in Wasserstein distance. $\square$

Density-level characterization - Assume that for each state s , the law $\eta_t(s)$ admits a density $p_{t,s}(z)$, and that the reward R_t admits a conditional density $p_R(r \mid s, a)$. Then the density of $(\mathcal{T}^\pi \eta_t)(s)$, denoted $p'_{t,s}(z)$, is given by

$$p'_{t,s}(z) = \gamma^{-1} \sum_{a \in \mathcal{A}} \pi(a \mid s) \int_{\mathbb{R}} p_R(r \mid s, a) \sum_{s' \in \mathcal{S}} P(s' \mid s, a) p_{t+1,s'}\left(\frac{z - r}{\gamma}\right) dr. \quad (25)$$

This representation makes explicit that the distributional Bellman operator acts as a *mixture-convolution*: the transition kernel induces a mixture over next-state densities, while the additive reward induces a convolution followed by the scaling $z \mapsto z/\gamma$.

Equivalently, in terms of cumulative distribution functions,

$$F_{(\mathcal{T}^\pi \eta_t)(s)}(z) = \mathbb{E}_\pi \left[F_{\eta_{t+1}(S_{t+1})} \left(\frac{z - R_t}{\gamma} \right) \mid S_t = s \right]. \quad (26)$$

This, however, poses a key difficulty - the Bellman operator now acts on probability measures rather than scalar value functions, which introduces nonlinearity, loss of contraction under standard norms, and difficulty in propagating moments such as variance. In the next section, we shall address the contraction properties and present the proposal in [1] of using the Wasserstein-2 distance as the contraction metric. Later, we will provide a detailed overview of the propagation of higher moments as in [10].

Contraction - The contraction behavior depends on the metric used. In particular, the operator generally is not a strict contraction in the sup norm on cumulative distribution functions, although it is a γ -contraction in the supremum p -Wasserstein metric, including the case $p = \infty$. This is the main conclusion in chapter 4 in [1], but before we proceed with the main theorem, consider the following preliminary definitions

ρ -homogeneity - A probability metric d is ρ -homogeneous if for random variables Z_1, Z_2 , and for the associated probability measures ν_1, ν_2 it satisfies

$$d(\gamma Z_1, \gamma Z_2) = \gamma^\rho d(Z_1, Z_2), \quad d((T_0 D_\gamma)_\# \nu_1, (T_0 D_\gamma)_\# \nu_2) = \gamma^\rho d(\nu_1, \nu_2) \quad (27)$$

Regularity - A probability metric d is *regular* if for random variables Z_1, Z_2 with associated probability measures ν_1, ν_2 , and for any random variable W independent of (Z_1, Z_2) , it satisfies

$$d(W + Z_1, W + Z_2) \leq d(Z_1, Z_2), \quad d((T_W)_\# \nu_1, (T_W)_\# \nu_2) \leq d(\nu_1, \nu_2). \quad (28)$$

p-convexity - Given $p \in [1, \infty)$, a probability metric d is *p-convex* if for $\alpha \in (0, 1)$ and probability measures $\nu_1, \nu_2, \nu'_1, \nu'_2$, it satisfies

$$d^p(\alpha \nu_1 + (1 - \alpha) \nu_2, \alpha \nu'_1 + (1 - \alpha) \nu'_2) \leq \alpha d^p(\nu_1, \nu'_1) + (1 - \alpha) d^p(\nu_2, \nu'_2). \quad (29)$$

Theorem 3 (Contraction of the distributional Bellman operator). *Assume the metric d_p on $\mathcal{P}(\mathbb{R})$ is p-convex and ρ -homogeneous, i.e.*

$$d_p((D_\gamma)_\# \mu, (D_\gamma)_\# \nu) = \gamma^\rho d_p(\mu, \nu), \quad \mu, \nu \in \mathcal{P}(\mathbb{R}).$$

Then for any two return-distribution functions $\eta_t, \eta'_t : \mathcal{S} \rightarrow \mathcal{P}(\mathbb{R})$,

$$\bar{d}_p(\mathcal{T}^\pi \eta_t, \mathcal{T}^\pi \eta'_t) \leq \gamma^\rho \bar{d}_p(\eta_{t+1}, \eta'_{t+1}), \quad (30)$$

where

$$\bar{d}_p(\eta, \eta') := \sup_{s \in \mathcal{S}} d_p(\eta(s), \eta'(s)).$$

Proof. Fix $s \in \mathcal{S}$. Conditioning on $A_t = a$, define the state-action variant

$$(\mathcal{T}^\pi \eta_t)(s, a) := \mathbb{E}_\pi[(T_{R_t} D_\gamma)_\# \eta_{t+1}(S_{t+1}) \mid S_t = s, A_t = a]. \quad (31)$$

Then

$$\begin{aligned} d_p((\mathcal{T}^\pi \eta_t)(s, a), (\mathcal{T}^\pi \eta'_t)(s, a)) &= d_p(\mathbb{E}_\pi[(T_{R_t} D_\gamma)_\# \eta_{t+1}(S_{t+1}) \mid S_t = s, A_t = a], \mathbb{E}_\pi[(T_{R_t} D_\gamma)_\# \eta'_{t+1}(S_{t+1}) \mid S_t = s, A_t = a]) \\ &\leq \sum_{s' \in \mathcal{S}} P(s' \mid s, a) d_p((T_{r(s,a)} D_\gamma)_\# \eta_{t+1}(s'), (T_{r(s,a)} D_\gamma)_\# \eta'_{t+1}(s')) \\ &\leq \gamma^\rho \sum_{s' \in \mathcal{S}} P(s' \mid s, a) d_p(\eta_{t+1}(s'), \eta'_{t+1}(s')) \\ &\leq \gamma^\rho \bar{d}_p(\eta_{t+1}, \eta'_{t+1}). \end{aligned} \quad (32)$$

Here the first inequality uses p -convexity, while the second uses translation invariance together with ρ -homogeneity under dilation. Averaging over policies will yield

$$(\mathcal{T}^\pi \eta_t)(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) (\mathcal{T}^\pi \eta_t)(s, a), \quad (\mathcal{T}^\pi \eta'_t)(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) (\mathcal{T}^\pi \eta'_t)(s, a).$$

Applying p -convexity once more,

$$\begin{aligned} d_p((\mathcal{T}^\pi \eta_t)(s), (\mathcal{T}^\pi \eta'_t)(s)) &\leq \sum_{a \in \mathcal{A}} \pi(a \mid s) d_p((\mathcal{T}^\pi \eta_t)(s, a), (\mathcal{T}^\pi \eta'_t)(s, a)) \\ &\leq \gamma^\rho \bar{d}_p(\eta_{t+1}, \eta'_{t+1}). \end{aligned} \quad (33)$$

Taking the supremum over $s \in \mathcal{S}$ yields

$$\bar{d}_p(\mathcal{T}^\pi \eta_t, \mathcal{T}^\pi \eta'_t) \leq \gamma^\rho \bar{d}_p(\eta_{t+1}, \eta'_{t+1}),$$

as required. $\square$

4 Variance Effects and Second-Order Structure

⁴ While the distributional Bellman equation characterizes the *full law* of the return, it is often useful to extract lower-order summaries such as the variance in order to quantify risk and dispersion. In particular, variance captures sensitivity to uncertainty that is not visible at the level of the mean. Hence, in this subsection, we provide an investigation of variance modeling [10] and its significance in the broader distributional reinforcement learning framework.

Theorem 4 (Variance Bellman Equation). *Using the one-step decomposition $G_t^\pi \stackrel{d}{=} r(S_t, A_t) + \gamma G_{t+1}^\pi$, with $A_t \sim \pi(\cdot | S_t)$ and $S_{t+1} \sim P(\cdot | S_t, A_t)$, we obtain*

$$\Sigma_t^\pi(s) = \gamma^2 \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} P(s'|s, a) \Sigma_{t+1}^\pi(s') + \theta_t^\pi(s) \iff \Sigma_t^\pi = \theta_t^\pi + \gamma^2 P^\pi \Sigma_{t+1}^\pi. \quad (34)$$

in component-wise and vector forms and the local variance contribution is

$$\theta_t^\pi(s) := \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} P(s'|s, a) (r(s, a) + \gamma V_{t+1}^\pi(s'))^2 - (V_t^\pi(s))^2. \quad (35)$$

Proof. For a fixed policy π , define the value function and variance function

$$V_t^\pi(s) := \mathbb{E}[G_t^\pi | S_t = s] \Rightarrow V_t^\pi(s) = \int_{\mathbb{R}} x \eta_t^\pi(s)(dx), \quad (36)$$

$$\Sigma_t^\pi(s) := \text{Var}(G_t^\pi | S_t = s) \Rightarrow \Sigma_t^\pi(s) = \int_{\mathbb{R}} (x - V_t^\pi(s))^2 \eta_t^\pi(s)(dx). \quad (37)$$

It is also true that the variance itself admits a recursive structure which can be exploited to formulate a dynamic program, under the standard assumptions of Markovian conditional dependence and time-homogeneity. The above recursion follows directly from the affine pushforward structure of the distributional Bellman operator

$$\eta_t^\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) (T_{r(s,a)} D_\gamma)_\# \eta_{t+1}^\pi(s'). \quad (38)$$

Under this transformation, moments evolve as

$$\mathbb{E}[r + \gamma X] = r + \gamma \mathbb{E}[X], \quad (39)$$

$$\text{Var}(r + \gamma X) = \gamma^2 \text{Var}(X). \quad (40)$$

Thus, variance propagation arises from the dilation operator D_γ , while $\theta_t^\pi(s)$ captures variability induced by stochastic transitions and policy mixing. $\square$

Law of Total Variance Form - Applying the law of total variance,

$$\text{Var}(X) = \text{Var}(\mathbb{E}[X | Y]) + \mathbb{E}[\text{Var}(X | Y)], \quad (41)$$

with $X = G_t^\pi$ and $Y = S_{t+1}$, we obtain $\mathbb{E}[G_t^\pi | S_{t+1} = s'] = r(s, a) + \gamma V_{t+1}^\pi(s')$ and $\text{Var}(G_t^\pi | S_{t+1} = s') = \gamma^2 \Sigma_{t+1}^\pi(s')$. Substituting yields

$$\Sigma_t^\pi(s) = \underbrace{\text{Var}_{s',a}(r(s, a) + \gamma V_{t+1}^\pi(s'))}_{\text{structural / transition variance}} + \underbrace{\gamma^2 \mathbb{E}_{s'}[\Sigma_{t+1}^\pi(s')]}_{\text{propagated variance}}. \quad (42)$$

The variance decomposition separates two fundamentally different sources of uncertainty

⁴For this section, one must ensure that the second-moment does indeed exist

- **Propagation of downstream uncertainty** - $\gamma^2 \mathbb{E}_{s'}[\Sigma_{t+1}^\pi(s')]$
- **One-step structural uncertainty** - $\theta_t^\pi(s)$

This decomposition mirrors the structure of the distributional Bellman operator and provides a second-order characterization of the return law. In particular, even when value functions coincide, differences in θ_t^π induce distinct risk profiles.

This recursion is not independent of the distributional Bellman equation, rather, it arises as a second-order projection of the measure-valued operator onto moments.

Remark 1 (Second-Order Bellman Operator). *The variance recursion defines a second-order Bellman operator*

$$\mathcal{B}_2^\pi \Sigma := \theta_t^\pi + \gamma^2 P^\pi \Sigma,$$

which depends on the first-order value function V_t^π . Thus, unlike classical dynamic programming, second-order structure is not closed and depends explicitly on lower-order moments.

4.1 Summary of Key Distinctions - Classical vs. Distributional RL

Object of study - Classical RL studies $V^\pi(x) \in \mathbb{R}$; distributional RL studies $\eta^\pi(x) \in \Delta(\mathbb{R})$. The latter is strictly richer.

Bellman equation - Classical uses ordinary equality; distributional requires $\stackrel{D}{=}$ to avoid incorrectly coupling reward random variables across time steps.

Sensitivity to dependence - Classical RL is insensitive to the joint distribution of $(R_t)_{t \geq 0}$ because expectation is linear. Distributional RL is sensitive: the shape of $\eta^\pi(x)$ depends on how rewards across time interact, and naive direct recursion produces wrong distributions.

Interpretation of γ - In classical RL, γ as a scaling factor and as a continuation probability are equivalent (same expected return). In distributional RL they are inequivalent: the two interpretations yield different return distributions.

Mathematical operations - Classical RL uses linear algebra on $\mathbb{R}^{|\mathcal{X}|}$. Distributional RL requires three new operations on $\Delta(\mathbb{R})^{|\mathcal{X}|}$: **mixing** (weighted combination of distributions), **scaling** (pushforward with $z \mapsto \gamma z$), and **translation** (pushforward with $z \mapsto r + z$), combined through the affine translation given as $T_R D_\gamma$

Generalization - The distributional Bellman equation implies the classical one by taking expectations. Classical RL is a special case obtained by projecting distributions to their first moment.

5 Example - Inventory Management Under a Risk Factor

In the standard inventory management problem, excess stock incurs holding cost and unmet demand incurs shortage cost. Here we consider a safety-critical variant motivated by the storage of hazardous materials such as chemicals or explosives. Let K denote the maximum safe storage capacity. If the end-of-day inventory exceeds this threshold, then the system is exposed to the possibility of catastrophic failure.

Assuming immediate delivery, the inventory evolves according to

$$S_{t+1} = S_t + A_t - W_t. \tag{43}$$

Let $x = s + a - w$ denote the end-of-day inventory. The one-step cost is defined by $c(s, a, w, \xi) = pa + h(s) + U \xi \mathbf{1}_{\{s > K\}}$ where $p > 0$ is the unit procurement cost,

$$h(s) = \begin{cases} c_h s, & x \geq 0, \\ -c_s s, & x < 0, \end{cases} \quad (44)$$

and $\xi \sim \text{Bernoulli}(q)$ models a catastrophic failure conditional on exceeding the safe storage threshold. Thus, whenever $x > K$, an additional cost U is incurred with probability q .

5.1 Classical Dynamic Programming Formulation

For a fixed policy π , define the discounted cumulative cost from time t onward by

$$C_t^\pi := \sum_{k=t}^T \gamma^{k-t} c(S_k, A_k, W_k, \xi_k), \quad (45)$$

where $\gamma \in [0, 1)$ is the discount factor. The associated value function is $V_t^\pi(s) := \mathbb{E}[C_t^\pi | S_t = s]$. Using the one-step decomposition of the cumulative cost, $C_t^\pi = c(S_t, A_t, W_t, \xi_t) + \gamma C_{t+1}^\pi$, the following recursive form is obtained

$$V_t^\pi(s) = \mathbb{E}^\pi[c(S_t, A_t, W_t, \xi_t) + \gamma C_{t+1}^\pi | S_t = s] = \sum_{a \in \mathcal{A}} \pi(a|s) \mathbb{E}_{W_t, \xi_t}[c(s, a, W_t, \xi_t) + \gamma V_{t+1}^\pi(s + a - W_t)] \quad (46)$$

Here the expectation is taken jointly over demand uncertainty and catastrophic-failure uncertainty. The optimal value function is therefore given by

$$V_t^*(s) = \min_{a \in \mathcal{A}} \mathbb{E}_{W_t, \xi_t}[c(s, a, W_t, \xi_t) + \gamma V_{t+1}^*(s + a - W_t)], \quad (47)$$

$$\pi_t^*(s) \in \arg \min_{a \in \mathcal{A}} \mathbb{E}_{W_t, \xi_t}[c(s, a, W_t, \xi_t) + \gamma V_{t+1}^*(s + a - W_t)]. \quad (48)$$

Since $\xi_t \sim \text{Bernoulli}(q)$, the catastrophic component contributes to the Bellman equation only through its expectation

$$\mathbb{E}[U \xi_t \mathbf{1}_{\{s+a-W_t > K\}}] = Uq \mathbb{P}(s + a - W_t > K). \quad (49)$$

Hence, the classical value function aggregates all uncertainty into a single scalar quantity, ignoring the tail probabilities which correspond to rare but severe catastrophic events or the high-risk high reward objectives. In particular, the resulting dynamic program selects actions by minimizing expected cumulative cost alone

$$V_t^*(s) = \min_{a \in \mathcal{A}} \left\{ pa + \mathbb{E}_{W_t}[h(s + a - W_t)] + Uq \mathbb{P}(s + a - W_t > K) + \gamma \mathbb{E}_{W_t}[V_{t+1}^*(s + a - W_t)] \right\}. \quad (50)$$

Thus, in the classical formulation, the effect of catastrophic failure is reduced to its average contribution. While this is sufficient for expected-cost minimization, it does not distinguish between policies that have similar mean cost but substantially different probabilities of rare, high-cost outcomes.

5.2 Distributional Dynamic Programming Formulation

For a fixed policy π , let

$$C_t^\pi := \sum_{k=t}^T \gamma^{k-t} c(S_k, A_k, W_k, \xi_k) \quad (51)$$

denote the discounted cumulative cost from time t onward. The one-step decomposition is $C_t^\pi \stackrel{d}{=} c(S_t, A_t, W_t, \xi_t) + \gamma C_{t+1}^\pi$. For each state s , let $\eta_t^\pi(s)$ denote the conditional law of the cumulative cost

$$\eta_t^\pi(s)(B) := \mathbb{P}(C_t^\pi \in B \mid S_t = s), \quad B \subseteq \mathbb{R} \text{ Borel.} \quad (52)$$

Then, conditioning on the first action and on the realizations of demand and catastrophic failure, the distributional Bellman equation is

$$\eta_t^\pi(s)(B) = \sum_{a \in \mathcal{A}} \pi(a|s) \mathbb{E}_{W_t, \xi_t} \left[\eta_{t+1}^\pi(s + a - W_t) \left(\frac{B - c(s, a, W_t, \xi_t)}{\gamma} \right) \right], \quad (53)$$

where $\frac{B - c(s, a, w, \xi)}{\gamma} := \{u \in \mathbb{R} : c(s, a, w, \xi) + \gamma u \in B\}$.

Equivalently, in pushforward form,

$$\eta_t^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \mathbb{E}_{W_t, \xi_t} \left[(T_{c(s, a, W_t, \xi_t)} D_\gamma)_\# \eta_{t+1}^\pi(s + a - W_t) \right]. \quad (54)$$

Thus, the law of cumulative cost is obtained by the following steps

1. First, translate the immediate next cost law by the realized one-step cost $c(s, a, W_t, \xi_t)$,
2. scale by the discount factor γ ,
3. average over the randomness in demand and catastrophic failure, as well as over policy choice.

The classical value function is recovered as the first moment of the law

$$V_t^\pi(s) = \int_{\mathbb{R}} x \eta_t^\pi(s)(dx), \quad (55)$$

while the conditional variance is recovered as the second central moment

$$\Sigma_t^\pi(s) = \int_{\mathbb{R}} (x - V_t^\pi(s))^2 \eta_t^\pi(s)(dx). \quad (56)$$

More generally, the law also provides direct access to tail-event probabilities such as

$$\mathbb{P}(C_t^\pi \geq \tau \mid S_t = s) = \eta_t^\pi(s)([\tau, \infty)), \quad (57)$$

which measure the probability of incurring cost above a critical threshold τ . In contrast with the classical formulation, the catastrophic component is not collapsed into its mean contribution. Indeed, the classical Bellman equation replaces the random term $U \xi_t \mathbf{1}_{\{s+a-W_t > K\}}$, and expectations produce $U q \mathbb{P}(s+a-W_t > K)$, thereby losing the distinction between a policy that incurs moderate costs frequently and one that incurs catastrophic costs only rarely. The distributional formulation preserves this distinction through the full law $\eta_t^\pi(s)$.

6 Learning Return Distributions

The distributional Bellman equation characterizes the collection of laws $\{\eta_t^\pi(s)\}_{t,s}$ recursively. In practice, however, these laws are not available analytically and must be estimated from sampled transitions or sampled trajectories.

Monte-Carlo Estimation - The learning problem generally concerns itself with the synthesis of optimal strategies without being informed of any structural properties of the skeletal MDP. The most direct approach is to estimate the law of the return from sampled realizations. For a trajectory initiated at time t from state $S_t = s$, the actual sampled return and its empirical estimate are

$$\hat{G}_t^{(i)}(s) = \sum_{n=t}^T \gamma^{n-t} R_n^{(i)}, \quad \hat{\eta}_t^\pi(s) = \frac{1}{N} \sum_{i=1}^N \delta_{\hat{G}_t^{(i)}(s)}. \quad (58)$$

respectively. This is a simple estimator with no bias due to raw sampling and consistent asymptotics, however, its main disadvantages, as with any Monte-Carlo based method are high variance and latent computation as one must wait for a full return rollout across a given trajectory.

Temporal-Difference Learning - TD-Learning is a good balance between dynamic programming and Monte-Carlo. The recursive structure when conditioned on $S_t = s$, $A_t = a$, and $S_{t+1} = s'$ is simply $G_t^\pi \stackrel{d}{=} r(s, a) + \gamma G_{t+1}^\pi$. Passing to laws yields the one-step target

$$(T_{r(s,a)} D_\gamma)_\# \eta_{t+1}^\pi(s'). \quad (59)$$

A stochastic approximation of the distributional Bellman recursion therefore takes the form

$$\eta_t(s) \leftarrow (1 - \alpha) \eta_t(s) + \alpha (T_{r(s,a)} D_\gamma)_\# \eta_{t+1}(s'). \quad (60)$$

Unlike the classical setting, however, the quantity being updated is now a probability law rather than a scalar, so a finite-dimensional representation is needed.

Categorical TD Learning - A common approximation is to represent each law $\eta_t^\pi(s)$ by a categorical distribution supported on a fixed grid. Fix atoms $z_1 < z_2 < \dots < z_K$, then the approximation becomes

$$\eta_t(s) \approx \sum_{i=1}^K p_{t,i}(s) \delta_{z_i}, \quad p_t(s) \in \Delta^K, \quad (61)$$

where $p_t(s) = (p_{t,1}(s), \dots, p_{t,K}(s))$ is a probability vector.

Distributional Target - Given a sampled transition (s, a, r, s') at time t with $Z'_{t+1} \sim \eta_{t+1}(s')$ The one-step distributional target is $Z_t^{\text{target}} = r + \gamma Z'_{t+1}$, and its law is $(T_r D_\gamma)_\# \eta_{t+1}(s')$. Since the support of Z_t^{target} will generally not coincide with the fixed atoms $\{z_i\}_{i=1}^K$, one introduces a projection operator

$$\Pi : \mathcal{P}(\mathbb{R}) \rightarrow \text{span}\{\delta_{z_1}, \dots, \delta_{z_K}\}, \quad (62)$$

$$\hat{\eta}_t^{\text{target}} = \Pi \left[(T_r D_\gamma)_\# \eta_{t+1}(s') \right]. \quad (63)$$

where $\hat{\eta}_t^{\text{target}}$ is the projected target law. The role of Π is to redistribute probability mass onto neighboring atoms so that the approximation remains within the chosen categorical family.

Update rule - Writing $\hat{p}_t^{\text{target}}$ for the probability vector associated with $\hat{\eta}_t^{\text{target}}$, the update is

$$p_t(s) \leftarrow (1 - \alpha) p_t(s) + \alpha \hat{p}_t^{\text{target}} \iff \eta_t(s) \leftarrow (1 - \alpha) \eta_t(s) + \alpha \Pi \left[(T_r D_\gamma)_\# \eta_{t+1}(s') \right] \quad (64)$$

where the right hand side is the the pushward perspective of the left hand side.

Categorical TD learning is therefore a projected stochastic approximation to the distributional Bellman recursion. Relative to classical TD learning, it replaces scalar bootstrapping by measure-valued bootstrapping

$$V_t^\pi(s) \leftarrow (1 - \alpha)V_t^\pi(s) + \alpha(r + \gamma V_{t+1}^\pi(s')) \quad \text{becomes} \quad \eta_t(s) \leftarrow (1 - \alpha)\eta_t(s) + \alpha \Pi \left[(T_r D_\gamma)_\# \eta_{t+1}(s') \right]. \quad (65)$$

Hence, classical TD propagates only the first moment, while categorical TD propagates an approximation of the full law.

Remark - The categorical approximation introduces projection error, but it yields a finite-dimensional parametrization of an otherwise infinite-dimensional object. As the number of atoms increases, the approximation becomes more expressive, while remaining computationally tractable.

7 Control - Policy Improvement

Thus far, the standard notions of performance evaluation has been introduced. Synthesizing a best policy requires further considerations - these have been discussed in this section. To state a summary of this section a-priori, note that there is little difference in policy improvement between the distributional and classical methods.

Risk-Neutral Control - The objective $J(\pi) = \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \right]$ is canonical to the distributional as well as the classical methods. A risk-neutral optimal policy is one which maximizes the objective $J(\pi)$. Typically this policy belongs to the broad class of history-dependent policies, however in lieu of the following proposition, one may restrict the optimal policy class to be a subset of deterministic, Markov policies.

Proposition 2. *Let π^* be a stationary, deterministic, Markov policy. Then*

$$J(\pi^*) \geq J(\pi), \forall \pi \in \pi_H \quad (66)$$

where the H subscript stands for history dependent-policies

Consequently, let π_{MS} denote the set of all stationary Markov policies to which the optimal policy belongs.

Value Iteration - The classical method used in control is to first **fix** $\pi \in \Pi_{MS}$, then

$$\textbf{Evaluation - } Q_t^\pi(s, a) = R_t(s, a) + \gamma \mathbb{E}[V_{t+1}^\pi(S_{t+1}) \mid s, a]$$

$$\textbf{Control - } Q_t^*(s, a) = R_t(s, a) + \gamma \mathbb{E}[V_{t+1}^*(S_{t+1}) \mid s, a]$$

$$\textbf{Value Iteration - } Q_{k+1}(s, a) = R(s, a) + \gamma \mathbb{E}[V_k(S')] \mid s, a]$$

$$V_{k+1}(s) = \max_a Q_{k+1}(s, a), \quad \pi_{k+1}(s) \in \arg \max_a Q_{k+1}(s, a)$$

The optimal policy for when $\lim_{k \rightarrow \infty}$ is the true optimal. If the algorithm runs for N iterations, then $V_{(N)}$ is a $\delta(N)$ approximate policy, where

$$\| \hat{V}_{(N)}^* - V^* \|_\infty \leq \delta(N) \quad (67)$$

The classical value iteration is contractive in the ℓ^∞ norm. Due to space-awareness the proof has been left inexplicit. However, one important note is that the control algorithms deal with the Bellman optimality operators, which are clearly non-linear unlike the policy evaluation Bellman operator which induces an affine transformation. Hence, contraction proofs require the use of a property of maximal differences

$$|\max_{a \in \mathcal{A}} f_1(s) - \max_{a \in \mathcal{A}} f_2(s)| \leq \max_{a \in \mathcal{A}} |f_1(s) - f_2(s)| \quad (68)$$

Q-Learning - The incremental Q-learning updates follow from the preceding discussion as

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(\{r + \gamma \max_{a \in \mathcal{A}} Q(S_{t+1}, a)\} - Q(s, a) \right) \quad (69)$$

where $r + \gamma \max_{a \in \mathcal{A}} Q(S_{t+1}, a)$ is the target. Q-learning has some potential negative implications due to the nested maximization term which induces maximization bias. Double Q-learning [12] proposes a straightforward extension of Q-learning to deal with maximization bias by proposing an alternating use of two Q-estimates in the incremental update expression. The standard methods described above are not sufficient for DistRL, particularly because the notion of *maximizing* a distribution does not make sense. Before dealing with how one may optimize policies on distributions, one needs to consider some practical implications of storing and utilizing an infinite dimensional object.

8 Projected Distributional Dynamic Programming

Why is this such a big deal?

Since none of the fixed-size representations are closed under $\mathcal{T}^\pi$, the algorithm must **project** the operator's output back into the representation after each application. This introduces a two-step procedure at each iteration

1. Apply the distributional Bellman operator: $\tilde{\eta}_{k+1} = \mathcal{T}^\pi \eta_k \in \mathcal{F}_E^\mathcal{X}$ (an unconstrained empirical distribution, potentially with many particles).
2. **Project** back into the fixed-size representation: $\eta_{k+1} = \Pi_{\mathcal{F}} \tilde{\eta}_{k+1} \in \mathcal{F}^\mathcal{X}$.

The combined operator is the **projected distributional Bellman operator**:

$$\Pi_{\mathcal{F}} \mathcal{T}^\pi : \mathcal{F}^\mathcal{X} \rightarrow \mathcal{F}^\mathcal{X}.$$

The choice of projection $\Pi_{\mathcal{F}}$ and the choice of representation $\mathcal{F}$ are coupled: a natural projection for the categorical representation is different from a natural projection for the quantile representation. **Projection for the categorical representation** - Given a distribution $\nu \in \mathcal{F}_E$ and fixed locations $\theta_1 < \dots < \theta_m$, the *categorical projection* Π_C distributes each particle's probability mass to its two nearest neighbors proportionally by distance. For a particle at location θ with probability p :

$$\Pi_C(\delta_\theta) = \frac{\theta_{i+1} - \theta}{\varsigma_m} \delta_{\theta_i} + \frac{\theta - \theta_i}{\varsigma_m} \delta_{\theta_{i+1}}, \quad \theta_i \leq \theta \leq \theta_{i+1}.$$

Particles falling outside $[\theta_1, \theta_m]$ are clipped to the nearest endpoint. This is the **projection step** used in the C51 algorithm (Chapter 3).

Projection for the quantile representation - Given a distribution $\nu \in \mathcal{F}_E$ and fixed number of particles m , the *quantile projection* Π_Q sets each location θ_i to be the $\hat{\tau}_i$ -quantile of ν , where $\hat{\tau}_i = (2i - 1)/(2m)$ are the *midpoint quantile levels*:

$$\Pi_Q(\nu) = \frac{1}{m} \sum_{i=1}^m \delta_{F_\nu^{-1}(\hat{\tau}_i)}.$$

This assigns equal probability to each particle and places particle i at the center of the i -th probability mass segment of ν . The midpoint quantile levels avoid placing particles at the exact boundaries of probability mass, which could introduce instability.

The key asymmetry is that categorical projection can misrepresent the shape of a sharply peaked distribution because atoms are locked to the grid, while quantile projection tracks the distribution's shape precisely but can't represent the range outside the data it has seen. The interactive demo above lets you see how increasing m makes both methods converge on the true shifted distribution.

Approximation error analysis - The quality of the projected operator depends on both how well the projection approximates the true Bellman update and how the error propagates across iterations. The chapter establishes approximation error bounds for both representations.

For the *categorical projection* with m atoms spanning $[V_{\min}, V_{\max}]$, the per-step projection error is bounded by the *grid resolution* $\varsigma_m/2$ in the w_1 Wasserstein metric:

$$w_1(\Pi_C \nu, \nu) \leq \frac{\varsigma_m}{2} \quad \forall \nu \text{ supported on } [V_{\min}, V_{\max}].$$

For the *quantile projection*, the per-step error in w_2 is bounded by the *optimal quantization error* of ν with m points, which decays as $m^{-1/2}$ for smooth distributions.

8.1 Convergence of Projected Dynamic Programming

A key concern is whether the projected operator $\Pi_{\mathcal{F}} \mathcal{T}^\pi$ still converges. Since the projection introduces an additional approximation at each step, the iterates $(\eta_k)_{k \geq 0}$ do not in general converge to η^π but to a *fixed point of the projected operator*, whose distance from η^π depends on the representational capacity.

The core technical result is that while $\Pi_{\mathcal{F}} \mathcal{T}^\pi$ is generally *not a contraction mapping* (the projection can in principle increase distances), it is a *non-expansion in the appropriate metric* for both the categorical and quantile representations. Combined with the γ -contraction of $\mathcal{T}^\pi$, this yields convergence of the iterates and a bound on the approximation error at convergence.

For the *categorical algorithm*, let η_C^* denote the fixed point of $\Pi_C \mathcal{T}^\pi$. Then

$$w_1(\eta_C^*(x), \eta^\pi(x)) \leq \frac{\varsigma_m}{2(1 - \gamma)} \quad \forall x \in \mathcal{X}.$$

The error is proportional to the grid resolution $\varsigma_m = (V_{\max} - V_{\min})/(m - 1)$ and inversely proportional to $1 - \gamma$, reflecting the accumulation of projection errors over an effective horizon of $1/(1 - \gamma)$ steps. Increasing m reduces the error at the cost of more parameters. For the *quantile algorithm*, the fixed point η_Q^* of $\Pi_Q \mathcal{T}^\pi$ satisfies a bound in the w_2 metric that depends on the complexity of η^π and the number of quantile particles m .

8.2 Summary of the Three Representations

Categorical representation ($\mathcal{F}_{C,m}$) Fixed atom locations, variable probabilities. Projection distributes mass to nearest neighbors. Naturally minimizes a KL divergence (cross-entropy) loss.

Compatible with gradient-based learning (used in C51). Error bound in w_1 : $O(\varsigma_m/(1 - \gamma))$. Not closed under $\mathcal{T}^\pi$ - requires projection.

Quantile representation ($\mathcal{F}_{Q,m}$) Variable atom locations, fixed equal probabilities $1/m$. Projection places atoms at empirical quantiles of the target. Naturally minimizes a quantile regression (pinball) loss. Error bound in w_2 depends on quantization complexity of η^π . Not closed under $\mathcal{T}^\pi$; requires projection.

m -particle representation ($\mathcal{F}_{E,m}$) Both locations and probabilities are free. Most expressive but hardest to work with algorithmically, as natural loss functions and projections are less well-defined.

Normal representation ($\mathcal{F}_N$) Only two parameters per state. Closed-form Bellman equations for mean and variance. Tractable but rarely expressive enough and collapses multimodality and non-Gaussianity.

Empirical representation ($\mathcal{F}_E$) Closed under $\mathcal{T}^\pi$ - can approximate η^π to arbitrary precision. But exponentially growing memory makes it computationally intractable beyond small examples.

8.3 Distributional Control

To establish the notion of optimization of policies for distributions, a greedy mapping needs to be defined.

$$\text{Let } \mathcal{G} : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{A})|_{MS} \quad (70)$$

where $\Delta(\mathcal{A})|_{MS}$ is the simplex over a subset of actions generated by a stationary Markov policy. That is, the definition may be alternatively stated loosely as $\mathcal{G} : \mathcal{S} \times \mathcal{A} \rightarrow \pi_{MS}$. Thus $\mathcal{G}(Q)$ is greedy with respect to Q .

$$\mathcal{G}(Q)(a|s) \geq 0 \implies Q(s, a) = \max_{a \in \mathcal{A}} Q(s, a) \quad (71)$$

For a return law given by $\eta \in \Delta_1(\mathbb{R})^{\mathcal{S} \times \mathcal{A}}$, the induced Q-value function is

$$Q_\eta(s, a) = \mathbb{E}_{Z \sim \eta(s, a)}[Z] = \mathcal{G}(\eta) \quad (72)$$

$$\mathcal{T}G = \mathcal{T}^{\mathcal{G}(\eta)}G \quad (73)$$

Remark - When the projected distributional Bellman equation preserves expectations, then the distributional and classical value iteration algorithms are the same.

Remark - As different distributions induce distinct greedy policies, the distributional bellman optimality operator is generally not unique.

Another difference between the classical and distributional control approaches is that the projection step must be explicitly factored into the value iteration algorithm. Hence, the value iteration over distributions consists of a projection step and a *greedification* step as follows

Let $\Pi_{\mathcal{F}}$ be the projection operator under a choice of representation class $\mathcal{F}$, then for any initial law $\eta \in \mathcal{F}^{\mathcal{S} \times \mathcal{A}}$

$$\eta_{k+1} = \Pi_{\mathcal{F}} \mathcal{T}^{\mathcal{G}(\eta_k)} \eta_k \quad (74)$$

Thus, the idea behind the distributional value iteration method is that at each step, a policy π is chosen such that it's greedy with respect to the expected value of the return law η , that is $\pi \in \mathcal{G}(\eta)$. Once, such a policy has been synthesized, it is used to compute the next estimate of the return

functional under a single step of distributional dynamic programming. The policy synthesis step makes use of the law-induced Q_{η_k} .

$$\textbf{Initial Condition} - \quad Q_0(s, a) = \mathbb{E}_{Z \sim \eta_0(s, a)}[Z], \quad (s, a) \in \mathcal{S} \times \mathcal{A} \quad (75)$$

$$\textbf{Recursion} - \quad Q_{k+1} = TQ_k \quad (76)$$

$$\text{Then} \quad Q_{\eta_k} = Q_k, \quad \forall k \geq 0 \quad (77)$$

$$\textbf{Greedy policy} - \quad \pi_k = \mathcal{G}(Q_k) \quad (78)$$

Theorem 5 (Convergence of Distributional Value Iteration under Greedy Policy). *Let $\mathcal{T}^G$ denote the distributional Bellman optimality operator induced by the greedy policy mapping $G(\eta)$, i.e.*

$$\eta_{k+1} = \mathcal{T}^{G(\eta_k)} \eta_k.$$

Suppose there exists a unique optimal policy $\pi^ \in \Pi_{MS}$ and assume the return distributions are such that the induced value functions are well-defined and bounded.*

Then, for any initial return-distribution function $\eta_0 : \mathcal{S} \rightarrow \Delta(\mathbb{R})$, the sequence $(\eta_k)_{k \geq 0}$ satisfies

$$\eta_k \xrightarrow[k \rightarrow \infty]{} \eta^{\pi^*}$$

with respect to the supremum p -Wasserstein metric $\bar{d}_p(\eta, \eta') := \sup_{s \in \mathcal{S}} d_p(\eta(s), \eta'(s))$.

Proof. Since the optimal policy π^* is unique, the associated action-gap is strictly positive. Define

$$\text{GAP}(Q^*) := \inf_{s \in \mathcal{S}} \left(Q^*(s, a^*(s)) - \max_{a \neq a^*(s)} Q^*(s, a) \right) > 0,$$

where $a^*(s)$ is the unique optimal action at state s . Let $\varepsilon = \frac{1}{2} \text{GAP}(Q^*)$.

From the classical contraction of the Bellman optimality operator in ℓ^∞ , we have that for sufficiently large k , say $k \geq K_\varepsilon$,

$$\|Q_{\eta_k} - Q^*\|_\infty < \varepsilon,$$

where $Q_{\eta_k}(s, a) := \mathbb{E}[Z]$, $Z \sim \eta_k(s, a)$.

Fix any state $s \in \mathcal{S}$ and let $a^*(s)$ be the optimal action. Then for any $a \neq a^*(s)$,

$$\begin{aligned} Q_{\eta_k}(s, a^*(s)) &\geq Q^*(s, a^*(s)) - \varepsilon \\ &\geq Q^*(s, a) + \text{GAP}(Q^*) - \varepsilon \\ &> Q_{\eta_k}(s, a) + \text{GAP}(Q^*) - 2\varepsilon \\ &= Q_{\eta_k}(s, a). \end{aligned}$$

Thus, for all $k \geq K_\varepsilon$, the greedy policy satisfies

$$G(\eta_k)(s) = \pi^*(s), \quad \forall s \in \mathcal{S},$$

i.e. $G(\eta_k) = \pi^*$.

Therefore, for $k \geq K_\varepsilon$, the iteration reduces to $\eta_{k+1} = \mathcal{T}^{\pi^*} \eta_k$. Since $\mathcal{T}^{\pi^*}$ is a γ^ρ -contraction under $\bar{d}_p$ (by Theorem 1), it follows that $\eta_k \rightarrow \eta^{\pi^*}$ as $k \rightarrow \infty$. The categorical distributional Q-learning method follows the following incremental update -

$$\eta(s, a) \leftarrow \eta(s, a) + \alpha \left(\sum_{a' \in \mathcal{A}} \mathcal{G}(\eta)(a' | s') \Pi_c(T_r D_\gamma)_{\#} \eta(s', a') \right) \Leftrightarrow Q_\eta(s, a) \leftarrow (1 - \alpha) Q_\eta(s, a) + \alpha \underbrace{\sum_{a' \in \mathcal{A}} \mathcal{G}(\eta)(a' | s') (r + \gamma \max_{a' \in \mathcal{A}} Q_{\eta'}(s, a'))}_{r + \gamma \max_{a' \in \mathcal{A}} Q_{\eta'}} \quad (79)$$

RHS in 79 follows LHS due to the fact that categorical methods are mean-preserving. As stated earlier, the greedy policies are not necessarily injective and may induce multiple distributional Bellman optimality operators. Under each, the dynamics are different and the analysis of such dynamics might be interesting. However, at this point, for the purpose of brevity, we refrain from this discussion and refer to [1]. $\square$

Experiment - Cliffwalking

The experimental framework evaluates classical temporal-difference control methods (SARSA, Expected SARSA, and Q-learning) alongside a categorical distributional Q-learning variant, following the previous discussion on policy control, on the CliffWalking-v1 environment. While classical methods learn scalar action-value functions the distributional approach instead propagates a full return distribution $\eta(s, a)$ over a fixed discrete support, using a projection operator to maintain tractability. Policy improvement in the distributional setting is performed via greedification with respect to the expected value of the learned distribution, ensuring consistency with risk-neutral control.

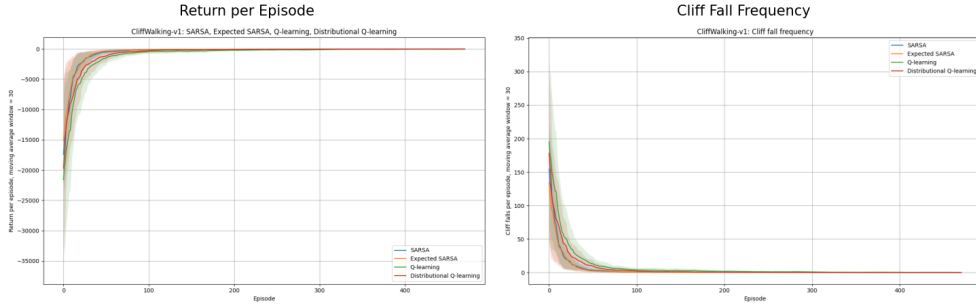


Figure 2: Mean Reward vs Episode and Mean Falls vs Episode

Figure 2 reaffirms the well-known result that SARSA and E-SARSA outperform Q-learning due to the maximization step which leads Q-learning suffering from risky decisions in pursuit of greedy strategies. Empirically, all methods exhibit convergence toward the optimal policy, as evidenced by increasing returns and a rapid decline in cliff-fall frequency. On-policy methods (SARSA and Expected SARSA) demonstrate smoother and safer early learning dynamics, incurring fewer catastrophic penalties due to their inherent risk-averse exploration behavior. In contrast, Q-learning initially suffers from higher variance and more frequent cliff falls, reflecting its off-policy maximization bias, but eventually converges to comparable performance. Distributional Q-Learning performs slightly worse than the other two. This is because it still suffers from the same maximization bias as classical Q-learning. However, the notable distinction here is that the distributional Q-learning algorithm performs better (learns faster) than classical Q-learning, albeit marginally, as indicated by the steeper curves. As the experiment has been conducted over 5 seeds, the information conveyed by the figure is fairly trustworthy. In lieu of that, it makes sense to conclude that making greedification be a function of the raw return law does help, but not to an extent that justifies absolute superiority for practical purposes, particularly because the training time of DistQL is significantly higher than that of Q-learning. Hence, to the pragmatic employer, this should not be viewed as distributional Q-learning being triumphant over classical Q-learning, but rather quite the opposite.

9 Distribution RL in Deep Learning setting

The distributional Bellman equation

$$Z(s, a) \stackrel{D}{=} R(s, a) + \gamma Z(s', \pi(s'))$$

preserves the full shape of the return distribution across Bellman updates, enabling the learning algorithm to distinguish actions whose return distributions differ in shape but share the same mean — a distinction that is invisible to traditional Q learning. Implicit Quantile Networks [3] realise this framework by learning the quantile function $Q_\tau(s, a)$ for $\tau \sim \mathcal{U}[0, 1]$ via the asymmetric quantile Huber loss

$$\rho_\tau(u) = |\tau - \mathbf{1}[u < 0]| \ell_\kappa(u),$$

which supervises the return distribution at every quantile level simultaneously, supplying a richer and lower-variance gradient signal than the single Huber loss of DQN and removing the need to pre-specify a fixed return support or quantile grid required by earlier distributional methods [1, 4]. Here u denotes the temporal-difference error between a target return and the predicted τ -quantile. The indicator term $\mathbf{1}[u < 0]$ creates an **asymmetric weighting**: over-estimation errors ($u < 0$) are penalized with weight $1 - \tau$, while under-estimation errors ($u > 0$) are penalized with weight τ , ensuring that each network output converges to the desired quantile rather than the mean. This asymmetry is multiplied by a Huber loss $\ell_\kappa(u)$, which behaves quadratically for small errors and linearly for large errors, providing robustness to outliers while retaining stable gradients. Its asymmetric weighting of over- and under-estimation errors ensures that each network output converges to a specific quantile of the return distribution, rather than collapsing toward the mean; a symmetric loss would instead minimize average error and therefore recover only expected value.

9.1 IQN and DQN for Bimodal problem

Environment Design and the Safe-Action Optimality. We study a two-action episodic investment environment in which an agent repeatedly chooses between a safe asset with deterministic per-step return $r_s = 0.05$ and a risky asset with stochastic return

$$r_r = \begin{cases} +0.12 & \text{with probability 0.9,} \\ -0.50 & \text{with probability 0.1,} \end{cases}$$

giving arithmetic expected return $\mathbb{E}[r_r] = 0.058 > r_s$. Per-step rewards are the raw return rates; no capital-scaled reward is used. Although the risky action has a higher arithmetic mean, terminal wealth is governed by the geometric mean growth factor $g = 1.12^{0.9} \times 0.5^{0.1} \approx 1.028$, compared with 1.05 for the safe asset. By the Kelly criterion, the safe action therefore maximises long-run compounded wealth despite its lower arithmetic mean, delivering a deterministic terminal capital of $\$100 \times 1.05^{50} \approx \$1,147$, whereas the risky action produces a heavily left-skewed episodic return distribution with geometric median $\approx \$512$ and a catastrophic lower tail from compounded -0.50 outcomes. A terminal bonus of $\log(C_T/C_0)/100$ is added to provide wealth signal but scaled down to avoid Itô leakage.

Kelly criterion. For the risky action with returns $+12\%$ (probability 0.9) and -50% (probability 0.1), the Kelly-optimal allocation f^* is obtained by maximizing the expected log-growth rate $0.9 \log(1 + 0.12f) + 0.1 \log(1 - 0.5f)$, yielding $f \approx 0.97$. Although the risky action has a higher arithmetic mean return, its Kelly-optimal allocation $f^* \approx 0.97$ achieves a geometric growth rate

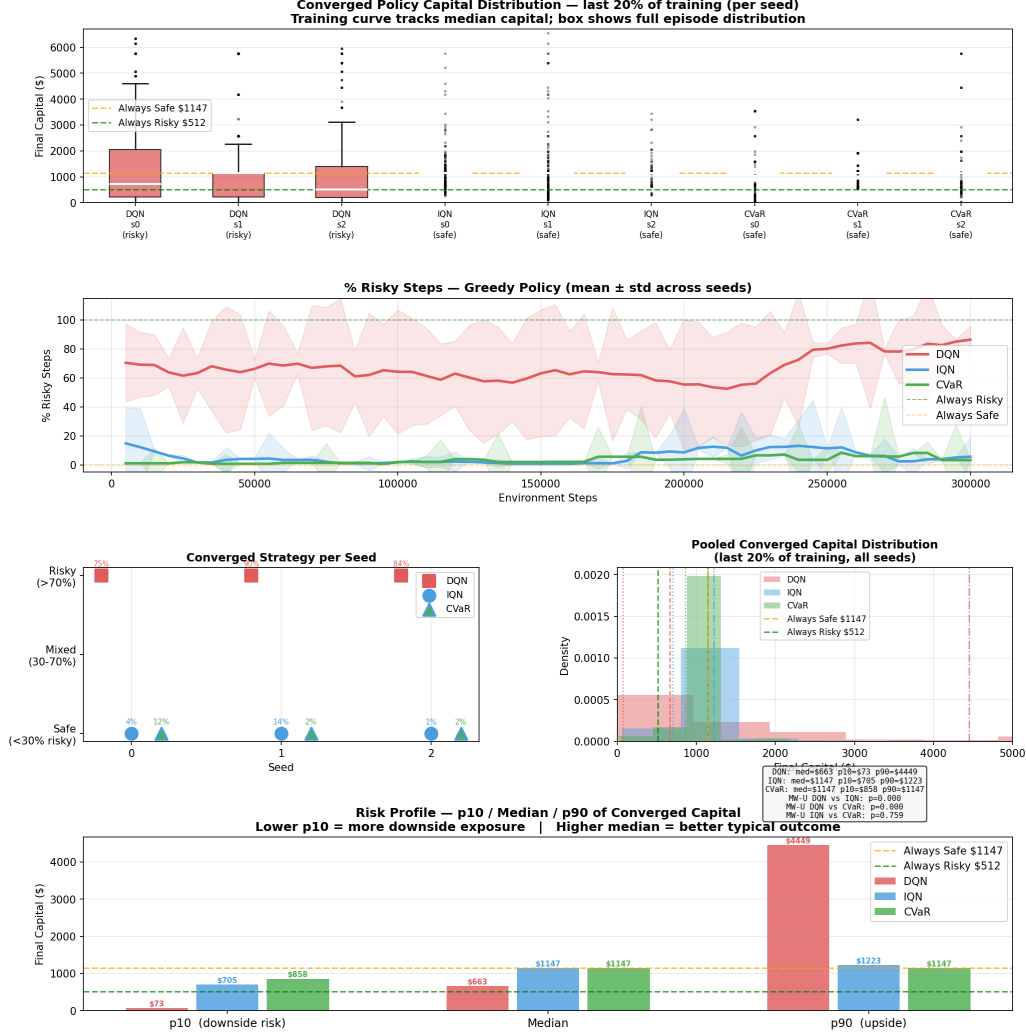


Figure 3: Performance of IQN and DQN.

of only about 3.3%, which is strictly dominated by the deterministic 5% return of the safe action — $g_{\text{safe}} = 1.05 > g_{\text{risky}}(f^*) \approx 1.0334$. Thus, even under optimal fractional betting, tail risk suppresses long-run compounded wealth, and the safe strategy remains optimal. This confirms that geometric growth—not expected value—governs rational decision-making over long horizons.

Empirical Results — Policy Divergence Across Algorithms. As shown in Figure 3, the three algorithms exhibit clearly separated converged policies across all seeds. DQN converges to a predominantly risky policy (≈ 75 – 90% risky steps), consistent with its objective of maximising arithmetic expected return ($\mathbb{E}[r_r] = 0.058 > r_s = 0.05$); this produces a heavily right-skewed capital distribution with median $\$663$, a catastrophic tenth percentile of $\$73$, and an inflated ninetieth percentile of $\$4,449$ driven by lucky all-win episodes. IQN converges to a near-pure safe policy across all three seeds ($\leq 14\%$ risky steps), concentrating terminal capital tightly around the always-safe baseline of $\$1,147$ ($p_{10} = \$705$, median = $\$1,147$, $p_{90} = \$1,223$), consistent with the implicit risk aversion induced by simultaneous multi-quantile gradient updates on catastrophic -0.50 outcomes described in the previous section. CVaR-RL converges to the safest policy of the three ($\leq 2\%$ risky steps across all seeds) and achieves the tightest capital distribution: median $\$1,147$, $p_{10} = \$858$, $p_{90} = \$1,147$, with no upside variance because the explicit CVaR objective at level $\alpha = 0.25$ systematically suppresses any action whose worst-quarter outcomes are inferior to the

safe alternative. Figures 4 and 5 visualise the training progress of the top-2 seeds per algorithm; the risky-step fraction for DQN trends upward throughout training while IQN and CVaR-RL stabilise near zero after $\approx 50,000$ steps.

Empirical Results — Risk Profile Comparison. The bottom panel of Figure 3 isolates the risk profiles of the three algorithms across three summary statistics of the converged capital distribution. On downside risk (p10), CVaR-RL (\$858) substantially outperforms both IQN (\$705) and DQN (\$73), confirming that the explicit CVaR objective provides stronger tail protection than IQN’s implicit conservatism and far stronger protection than DQN’s mean-greedy policy. On median capital, IQN and CVaR-RL are identical (\$1,147, matching the always-safe deterministic baseline) while DQN’s median (\$663) falls well below both the safe baseline and the always-risky geometric median of \$512, reflecting the compounded damage of periodic -0.50 outcomes under a persistently risky policy. On upside (p90), DQN’s distribution extends to \$4,449 — nearly four times the safe baseline — while IQN (\$1,223) and CVaR-RL (\$1,147) sacrifice this right-tail gain entirely in exchange for downside protection. A pairwise Mann-Whitney U test confirms that DQN’s capital distribution is significantly different from both IQN and CVaR-RL ($p < 0.001$), while IQN and CVaR-RL are not significantly different at the distribution level ($p = 0.759$), indicating that both distributional methods converge to the same safe-optimal terminal capital despite their mechanistic differences: IQN through implicit gradient-asymmetry conservatism and CVaR-RL through an explicit risk objective encoded in the Bellman target.

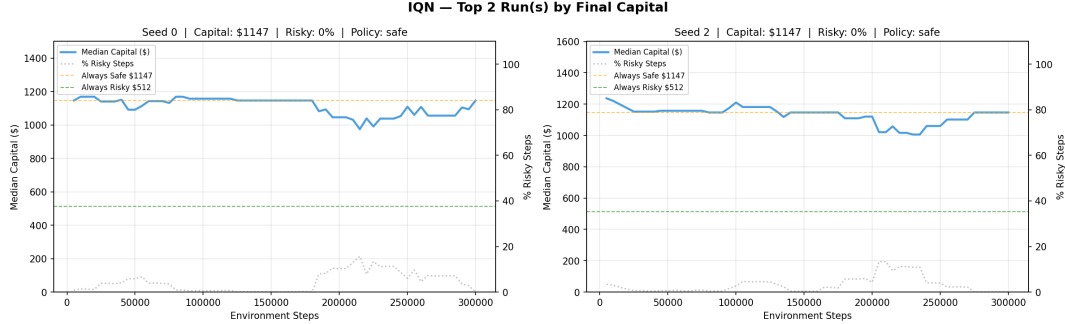


Figure 4: Training progress and benchmark of IQN.

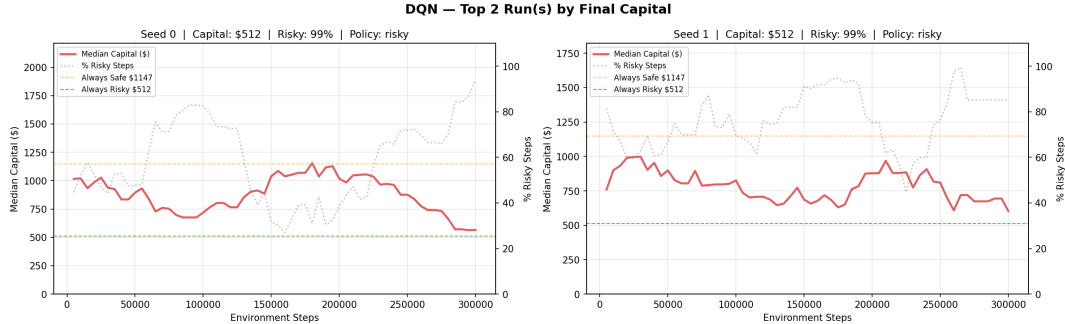


Figure 5: Training progress and benchmark of DQN.

On the Asymptotic Status of IQN’s Safe Preference. IQN prioritises risky action despite expected return $\mathbb{E}[r_r] = 0.058 > r_s$. A natural question is whether the observed safe preference of IQN is a transient training artefact that would vanish with sufficient training, or a stable property of the learned policy. To answer this precisely, it is necessary to first correct a common conflation: fixed $\varepsilon > 0$ does *not* create an asymptotic behavioural difference between DQN and IQN. Both

algorithms use an off-policy max-greedy Bellman target,

$$y = r + \gamma \max_{a'} Q(s', a'),$$

whose max operator is independent of the behaviour policy’s ε . This is precisely what makes Q-learning off-policy: it converges to Q^* regardless of ε , unlike SARSA which converges to the optimal ε -greedy policy [11]. Since DQN and IQN share the same target structure, fixed ε affects them identically and cannot explain the divergence. The true explanation involves three mechanisms that operate on different timescales.

Mechanism 1 — Quantile crossing bias (structural but diminishing). With finite $N = 16$ quantiles and a heavy-tailed risky return distribution, the independently estimated quantile levels cross [15]: lower- τ estimates temporarily exceed higher- τ estimates, introducing a systematic downward bias in the empirical mean $\hat{Q}(s, \text{risky}) = \frac{1}{N} \sum_{i=1}^N Q_{\tau_i}(s, \text{risky})$ relative to the true expectation $\mathbb{E}[Z(s, \text{risky})]$. This bias is provably worst for small N and heavy tails, and diminishes as estimates stabilise, but with $N = 16$ fixed a residual downward distortion from the -0.50 tail may persist. DQN maintains a single scalar estimate with no crossing pathology.

Mechanism 2 — Policy lock-in via sparse visitation [5] (potential, partially contested). Once the training trajectories push IQN toward the safe policy, the risky action is visited only at the exploration rate $\varepsilon = 0.02$, yielding ≈ 5 risky transitions per batch of $B = 256$. One might expect that $N = 16$ independent quantile regression targets, each fitted from these 5 samples, would produce higher-variance estimates than DQN’s single scalar — creating a self-reinforcing loop:

$$\text{IQN} \xrightarrow{\text{early bias}} \pi_{\text{safe}} \xrightarrow{\text{sparse risky visits}} \hat{Q}(s, \text{risky}) \text{ underestimated} \rightarrow \pi_{\text{safe}} \rightarrow \dots$$

However, this argument is contested on two grounds. First, each of the 5 risky samples contributes $N = 16$ quantile loss terms, giving 80 gradient updates to $\hat{Q}(\text{risky})$ versus DQN’s 5 — IQN is not obviously data-hungrier per batch. Second, over 300K steps the buffer accumulates $\sim 24,000$ risky transitions, sufficient in principle to estimate a two-outcome distribution at any fixed N . The feedback loop nonetheless remains a *plausible contributing factor*: the continued action of mechanisms 1 and 2 under sparse risky data refreshes the downward bias at each update, slowing correction of the quantile mean below the safe threshold and extending the lock-in well beyond what scalar averaging would permit.

Mechanism 3 — Non-uniqueness of the distributional optimality fixed point. The deepest and most potentially principled explanation comes directly from the theory of distributional Bellman operators. [1] prove that while the scalar Bellman optimality operator is a γ -contraction with a *unique* fixed point Q^* — ensuring that Q-learning and DQN converge to the same solution regardless of initialisation — the distributional Bellman optimality operator $\mathcal{T}$ is *not a contraction* and does not always possess a fixed point: although the iterates satisfy $\|\mathbb{E}[Z_k] - Q^*\|_\infty \rightarrow 0$ exponentially (the means converge correctly), the full distribution Z_k itself has no unique attractor under $\mathcal{T}$. Under neural function approximation the projected distributional operator can therefore stabilise at different solutions depending on the initialisation and gradient trajectory. Mechanisms 1 and 2 steer IQN’s early trajectory toward the safe policy; the non-uniqueness of the distributional fixed point then means there is no unique corrective force to pull it back — DQN experiences no such ambiguity because Q^* is the sole fixed point.

Summary The phenomenon is therefore neither a finite-sample nor an exploration artefact but a structural consequence of the distributional control setting, and one that can be deliberately exploited: by tuning the number of quantiles N , practitioners effectively adjust how strongly this implicit conservatism manifests — smaller N amplifies the gradient asymmetry and crossing bias, yielding stronger safe preference, while larger N reduces both effects and moves IQN closer to the risk-neutral behaviour of DQN, without any explicit modification of the risk objective.

9.2 Distributional RL in Application

Distributional Reinforcement Learning in High-Stakes Applications. In domains with asymmetric consequences, modelling the full return distribution $Z^\pi(s, a)$ rather than its expectation $Q^\pi(s, a) = \mathbb{E}[Z^\pi(s, a)]$ constitutes a qualitative shift: standard methods collapse all information about variance, skewness, and tail risk, whereas distributional RL preserves it and makes it accessible to downstream risk measures. Given a learned quantile function $Z_\tau(s, a)$, risk-sensitive policies can be obtained via distortion measures such as

$$\text{CVaR}_\alpha[Z^\pi(s, a)] = \frac{1}{\alpha} \int_0^\alpha Z_\tau(s, a) d\tau, \quad (80)$$

which captures worst-case outcomes and cannot be recovered from point estimates.

This capability underpins applications where failures carry significant cost. In autonomous racing, [14] used a distributional actor-critic (QR-SAC) to train Gran Turismo Sophy, achieving superhuman performance by stabilising value estimates in regimes dominated by crash-induced tail risk. In robotics, [9] showed that Distributional PPO (DPPO) combined with CVaR optimisation reduces catastrophic falls in quadrupedal locomotion compared to standard PPO, which optimises only expected return. In finance, distributional RL aligns naturally with established risk frameworks (e.g. VaR and CVaR), making it suitable for portfolio optimisation where tail events dominate long-term performance [2]. More broadly, [7] highlight applications across neuroscience, healthcare, and algorithm design, noting that distributional targets provide richer learning signals while enabling explicit risk sensitivity.

A recent breakthrough — Iterated IQN (i-IQN) [13] further amplifies the distributional advantages of distributional RL by applying multiple consecutive Bellman updates within each gradient step, learning a chain of action-value functions where each serves as the bootstrap target for the next. Whereas standard temporal-difference learning minimizes only a single one-step Bellman error and accumulates approximation error across iterations, i-QN minimizes all intermediate Bellman errors simultaneously, tightening error-propagation bounds and accelerating convergence. When instantiated with IQN, this iterated scheme retains full distributional modeling while compressing the effective planning horizon per update, making distributional representation and iterated Bellman updates complementary rather than redundant and yielding the strongest empirical performance reported.

10 Conclusion

This report presented a comprehensive analysis of distributional reinforcement learning from both theoretical and applied perspectives. The main contributions may be summarized as follows

- **Significance of Higher-Order Structure** - The impact of higher moments in sequential decision-making and how the distributional framework [1] and the earlier [10] capture it's effects were discussed. The variance Bellman equation demonstrated that second-order quantities are not closed under the Bellman operator and depend explicitly on the value function, highlighting the limitations of moment-based approaches and the need for richer representations.
- **Representation and Distributional Bellman Operators - Shift** - The framework transitions from scalar value functions V^π to return distributions $\eta^\pi(s) \in \Delta(\mathbb{R})$, thereby lifting dynamic programming from finite-dimensional vector spaces to infinite-dimensional spaces of

probability measures. The distributional Bellman equation was derived in random variable, CDF, and measure-theoretic forms. The classical Bellman equation was shown to arise as a first-moment projection of this more general formulation. The distributional Bellman operator was analyzed, revealing that contraction properties are not preserved under standard norms, but can be recovered under suitable probability metrics such as Wasserstein distances.

- **Computational Challenges** - The infinite-dimensional nature of return distributions necessitates finite representations. Projected distributional dynamic programming introduces a trade-off between expressivity, tractability, and approximation error, with categorical and quantile methods providing practical solutions.
- **Illustrative examples** - The inventory management example showed how distributional methods capture tail risks that are obscured under expectation-based formulations. The deep reinforcement learning experiments demonstrated that distributional methods can induce qualitatively different policies in the presence of asymmetric or heavy-tailed returns.

11 Critical Analysis

Distributional reinforcement learning addresses a fundamental limitation of classical approaches by modeling the full return distribution rather than only its expectation. Sobel (1982) showed that higher-order statistics, such as variance, play a role in analyzing returns. Distributional reinforcement learning generalizes this perspective by modeling the entire distribution, which, when moments exist, determines all moments, whereas the converse is not generally true. This provides a richer representation of uncertainty but introduces additional challenges and trade-offs, discussed below.

Note on Assumptions - The assumptions made within this framework are fairly mild and not overly restrictive. In fact, [1] and the related works on DistRL provides a generalization and further relaxations of the axiomatics associated with parallel ideas.

1. The main idea behind distributional reinforcement learning is to be able to capture all higher-moments including the mean. So, trivially this also captures the significance of variance as depicted in [10]. Due to its richness of information representation, it is generally counterproductive if the existence of the higher moments it is supposed to encapsulate is not guaranteed. [10] focuses on the second moment (variance) of the return, providing a recursive formulation analogous to the Bellman equation. However, higher-order moments are not considered, and in general, moments alone do not uniquely characterize the return distribution because two dissimilar distributions may have the same sequence of moments. ⁵
2. Algorithmic implementation of distributional reinforcement learning is not trivial. This is particularly due to the fact that return distributions are infinite dimensional. Learning such representations are typically not cheap.
3. In the summary provided earlier, on empirical representations, there is a clear three-way trade-off between expressivity, closedness, and tractability. The empirical representation maximizes the first two but sacrifices the third. The normal distribution maximizes tractability but sacrifices expressivity. The fixed-size empirical representations — particularly the categorical and quantile representations — occupy a carefully engineered middle ground, using a projection

⁵[1] The p -Wasserstein distance provides a metric on probability distributions that captures differences in their geometry and spread, rather than relying solely on moments.

step to enforce tractability while retaining sufficient expressivity to approximate real return distributions.

4. Distributional control requires the synthesis of optimal/greedy policies from expected values of return laws. For a framework that highlights scalar expected values as a major downside of the classical method, this is a sharp contrast - the decision-making step uses an expectation largely similar to classical Q-learning and under some assumptions, the distributional value iteration algorithm is the exact same as the standard value iteration.

Remark - The impact of this method has been primarily displayed through it's ability to capture risk, both here and in [1]. A natural question that might arise is how this performs better than risk-sensitive utilities. The first benefit offered by distributional reinforcement learning is that it is dynamic and hence enables one to formulate a dynamic program on a given problem, unlike static risk measures like CVaR. The follow-up to that response is the provision of insights into it's relative performance against dynamic risk measures such as entropic risk. To provide a formal and rigorous rebuttal to this reservation, one would need to consider something technically involved and derail oneself from the main topic, but concisely speaking, the risk-sensitive methods essentially modify the objective of a decision-making problem, while distributional RL changes it's representation. The immediate implication derived from this observation is the lack of a canonical dynamic risk measure. There is no general objective that is able to capture the risks associated with a variety of decision-making problems - finance, game-playing, control engineering - all use different notions of risk, e.g control engineers use ideas from robust control, financial analysts use CVaR, etc. So to define such a measure, one needs to restrict themselves to a specific functional form which maybe misspecified, offer limited flexibility, etc. In contrast, DistRL provides a general space where any notion of risk utility may be realized and used.

The link to all the experiments are [6]

Note on Length - We apologize for our inability to compress this document beyond its present state. Our primary reference, [1], is a book and not a paper. The suggestions that were provided to us <https://mycourses2.mcgill.ca/d21/1e/839143/discussions/topics/344754/View> alongside the additional suggestions on the day of the presentation requires a thorough review of the first 7 chapters of the book. Despite our best efforts, it is practically impossible to cut short on the document without losing some important ideas that we want to convey. Ideally, we would have liked to provide further proofs and derivations of the projected dynamic programming section, which now, has been kept mostly descriptive.

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